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Course code: DSA0602

Course name: Data Handling and Visualization

CLASS ACTIVITY 15.07.2026

1. Smart City Air Quality Monitoring:

CODE:

```
library(ggplot2)

air <- data.frame(
  Temperature = c(25, 28, 30, 32, 35),
  `PM2.5` = c(40, 55, 70, 90, 110)
)

ggplot(air, aes(x = Temperature, y = `PM2.5`, color = `PM2.5`)) +
  geom_point(size = 4) +
  scale_color_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "PM2.5 Hotspots (Sequential Color Scale)",
    x = "Temperature (°C)",
    y = "PM2.5",
    color = "PM2.5"
```

) +

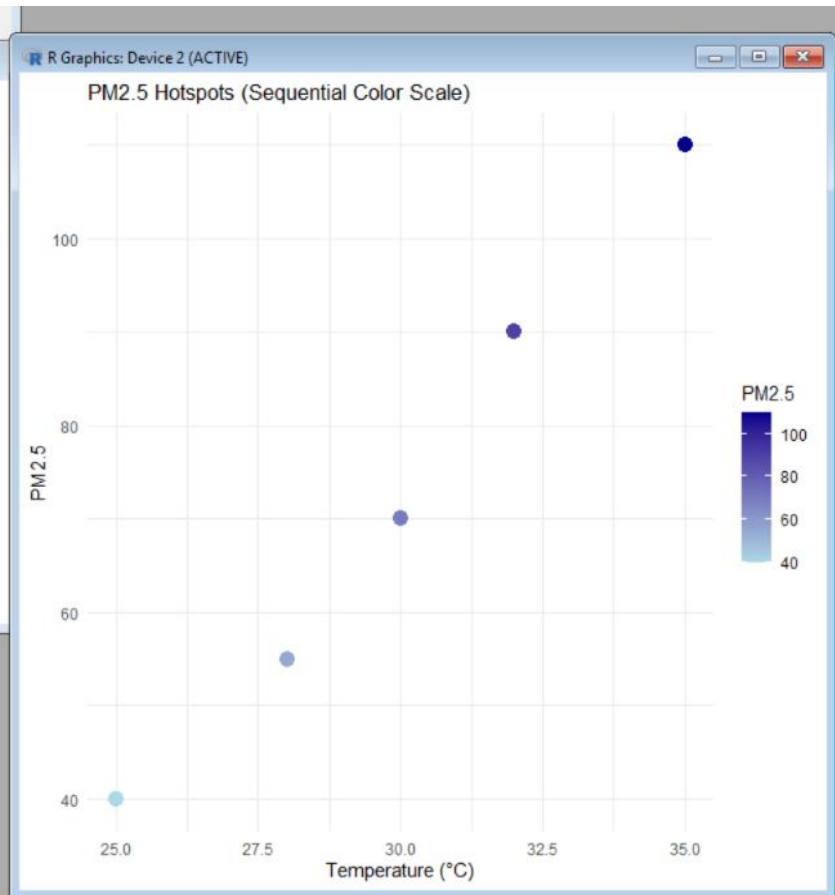
theme_minimal()

Wind Speed"

```
library(ggplot2)

air <- data.frame(
  Temperature = c(25, 28, 30, 32, 35),
  `PM2.5` = c(40, 55, 70, 90, 110)
)

ggplot(air, aes(x = Temperature, y = `PM2.5`, color = `PM2.5`)) +
  geom_point(size = 4) +
  scale_color_gradient(
    low = "lightblue",
    high = "darkblue"
  ) +
  labs(
    title = "PM2.5 Hotspots (Sequential Color Scale)",
    x = "Temperature (°C)",
    y = "PM2.5",
    color = "PM2.5"
  ) +
  theme_minimal()
```



2. Earthquake Risk Analysis:

CODE:

```
library(ggplot2)
```

```
air <- data.frame(
```

```
  Temperature = c(25, 28, 30, 32, 35),
```

```
  `PM2.5` = c(40, 55, 70, 90, 110)
```

```
)
```

```
ggplot(air, aes(x = Temperature, y = `PM2.5`, color = `PM2.5`)) +
```

```
  geom_point(size = 4) +
```

```
  scale_color_gradient2(
```

```

low = "blue",

mid = "yellow",

high = "red",

midpoint = mean(air$`PM2.5`)

) +

labs(

  title = "PM2.5 Hotspots (Diverging Color Scale)",

  x = "Temperature (°C)",

  y = "PM2.5",

  color = "PM2.5"

) +

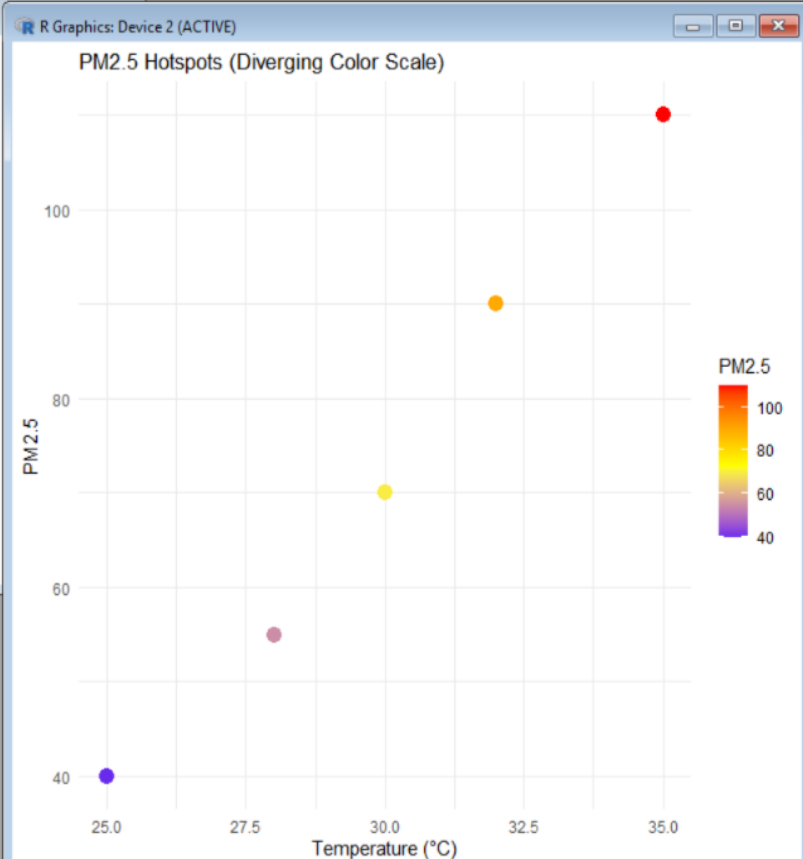
theme_minimal()

```

```

library(ggplot2)
air <- data.frame(
  Temperature = c(25, 28, 30, 32, 35),
  `PM2.5` = c(40, 55, 70, 90, 110)
)
ggplot(air, aes(x = Temperature, y = `PM2.5`, color = `PM2.5`)) +
  geom_point(size = 4) +
  scale_color_gradient2(
    low = "blue",
    mid = "yellow",
    high = "red",
    midpoint = mean(air$`PM2.5`)
  ) +
  labs(
    title = "PM2.5 Hotspots (Diverging Color Scale)",
    x = "Temperature (°C)",
    y = "PM2.5",
    color = "PM2.5"
  ) +
  theme_minimal()

```



3. Wind Direction Analysis for Airport Operations:

Visualization 1: Polar Coordinate (Wind Rose)

```
library(ggplot2)

wind <- data.frame(
  Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
  Speed = c(18, 25, 15, 10, 8, 12, 20, 16))

ggplot(wind, aes(x = Direction, y = Speed, fill = Speed)) +
  geom_bar(stat = "identity", color = "black") +
  coord_polar() +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(
    title = "Wind Rose - Airport Wind Direction",
    x = "Wind Direction",
    y = "Wind Speed"
  ) +
  theme_minimal()
```

Wind Rose - Airport Wind Direction

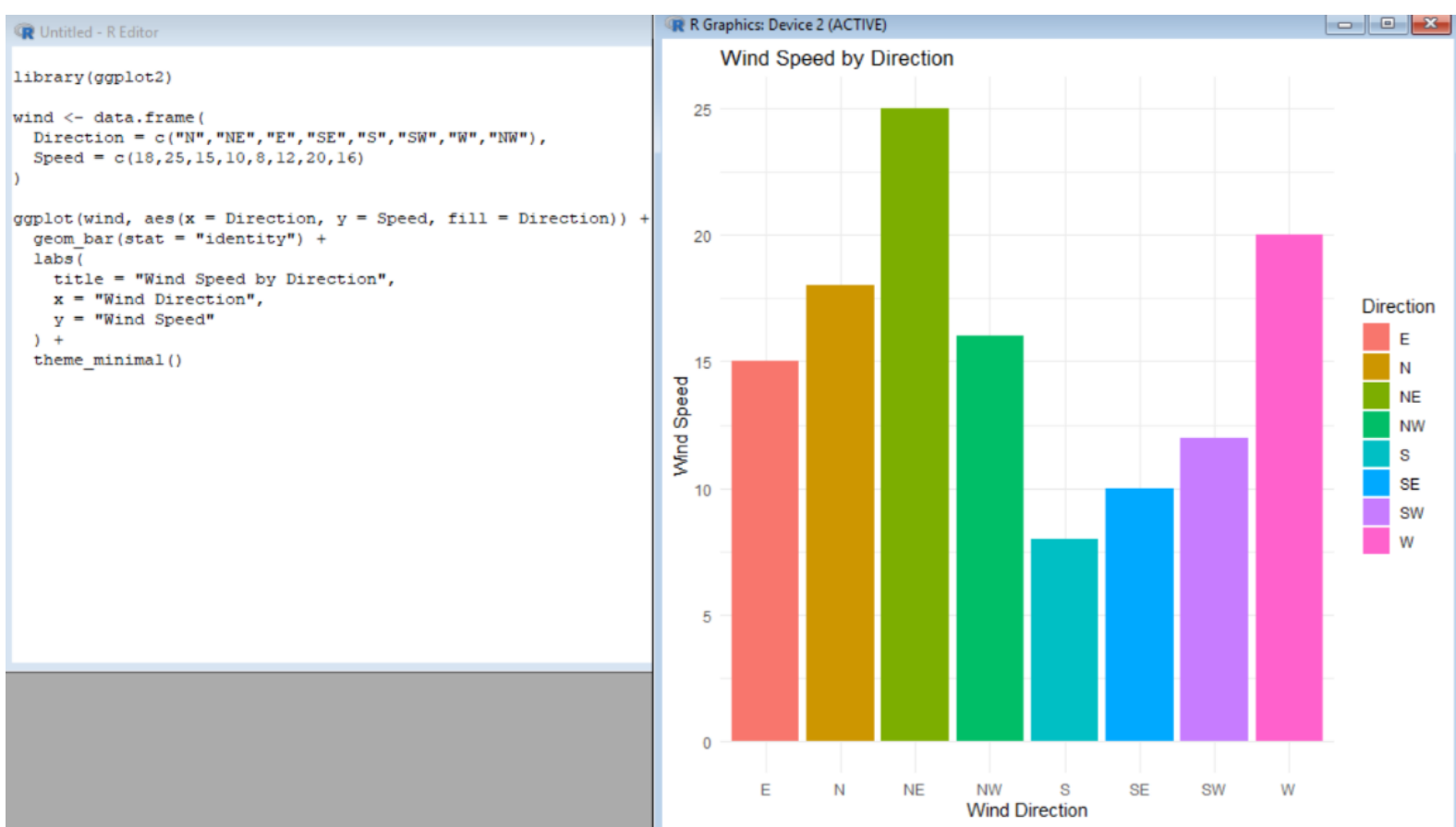
Wind Speed

Wind Direction

Speed

24
20
16
12
8

```
theme_minimal()
```



4. Hospital Patient Monitoring Dashboard:

CODE:

```
library(ggplot2)

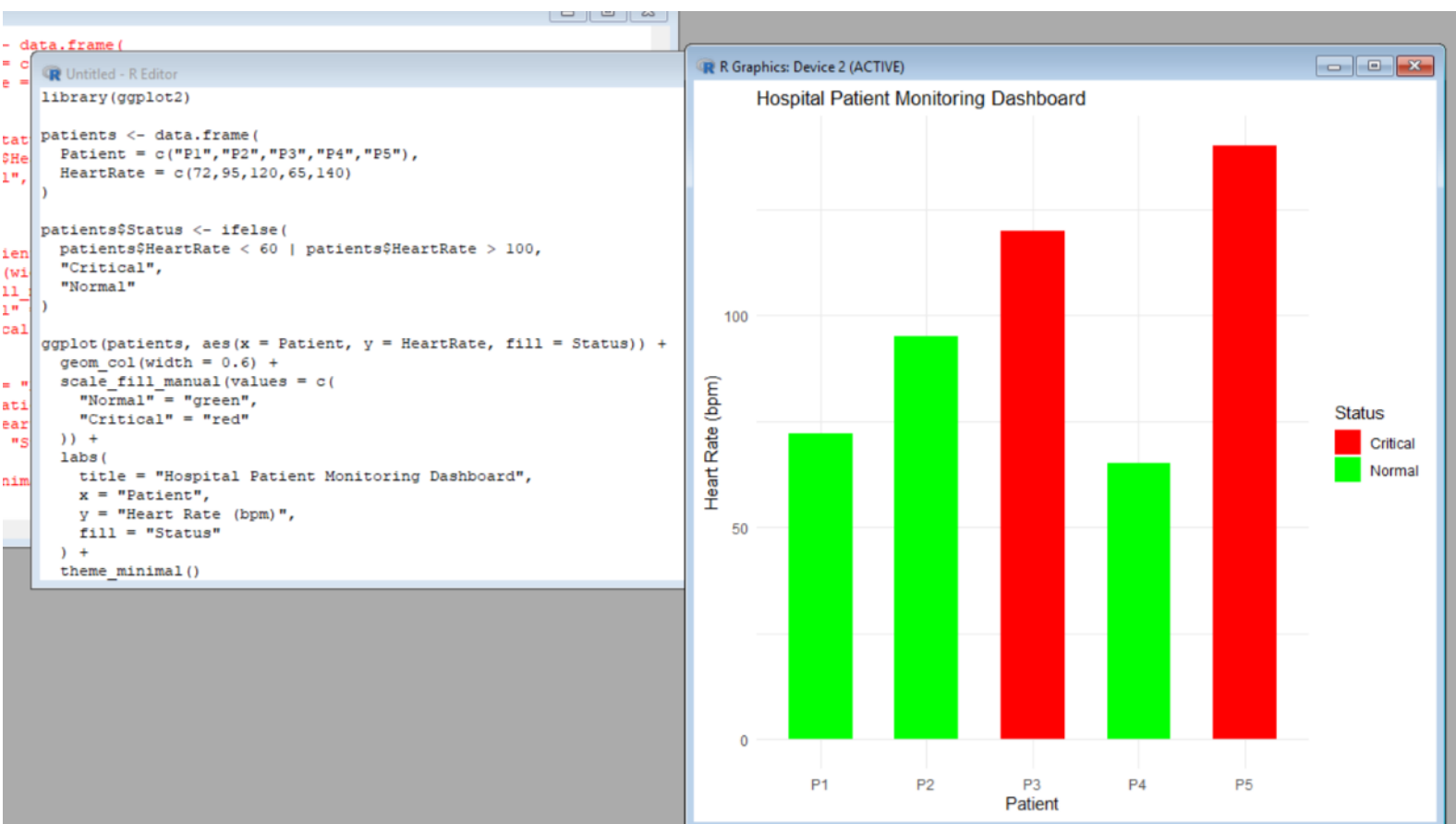
patients <- data.frame(
  Patient = c("P1", "P2", "P3", "P4", "P5"),
  HeartRate = c(72, 95, 120, 65, 140)

  patients$Status <- ifelse(
    patients$HeartRate < 60 | patients$HeartRate > 100,
    "Critical",
    "Normal"
  )
)
```

```

ggplot(patients, aes(x = Patient, y = HeartRate, fill = Status)) +
  geom_col(width = 0.6) +
  scale_fill_manual(values = c(
    "Normal" = "green",
    "Critical" = "red"
  )) +
  labs(
    title = "Hospital Patient Monitoring Dashboard",
    x = "Patient",
    y = "Heart Rate (bpm)",
    fill = "Status"
  ) +
  theme_minimal()

```



5. Global Climate Change Study:

CODE

Palette 1 – Blue to Red

```
library(ggplot2)

climate <- data.frame(
  Country = paste("C",1:10,sep=""),
  Year = rep(c(2000,2005,2010,2015,2020),2),
  Anomaly = c(-1.5,-0.8,-0.2,0.5,1.2,-1.0,-0.5,0.3,0.8,1.6)
)

ggplot(climate,aes(x=Year,y=Country,fill=Anomaly))+
  geom_tile()+
  scale_fill_gradient2(low="blue",mid="white",high="red",midpoint=0)+
  labs(title="Temperature Anomalies (Blue-White-Red)")+
  theme_minimal()
```



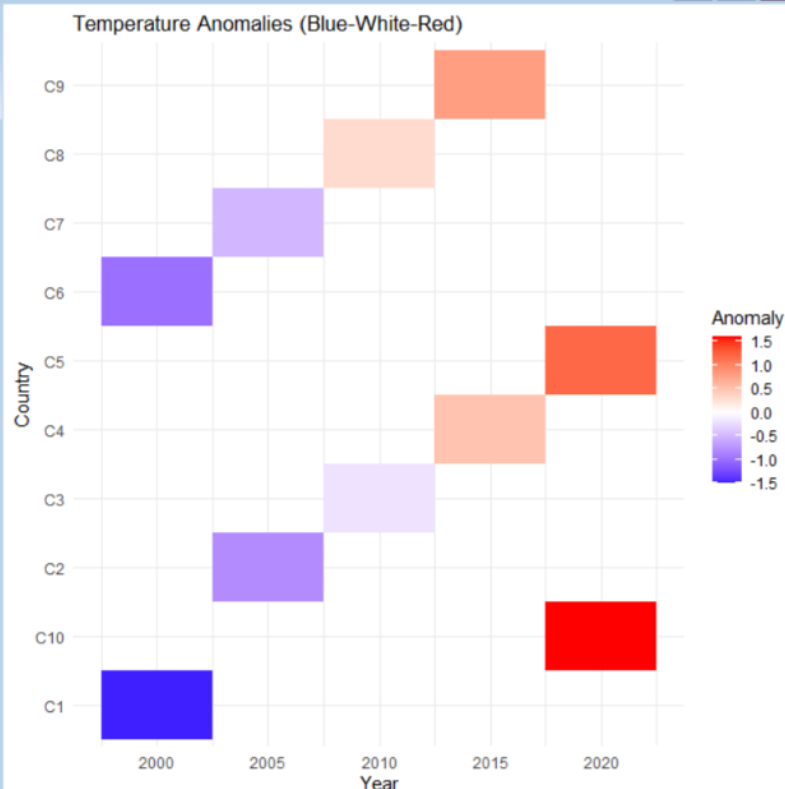
```
library(values = c(
```

```
Untitled - R Editor
```

```
library(ggplot2)
```

```
climate <- data.frame(  
  Country = paste("C",1:10,sep=""),  
  Year = rep(c(2000,2005,2010,2015,2020),2),  
  Anomaly = c(-1.5,-0.8,-0.2,0.5,1.2,-1.0,-0.5,0.3,0.8,1.6)  
)  
  
ggplot(climate,aes(x=Year,y=Country,fill=Anomaly))+  
  geom_tile()+  
  scale_fill_gradient2(low="blue",mid="white",high="red",midpoint=0)+  
  labs(title="Temperature Anomalies (Blue-White-Red)")+  
  theme_minimal()
```

```
R Graphics: Device 2 (ACTIVE)
```



Palette 2 – Green to Purple

```
library(ggplot2)
```

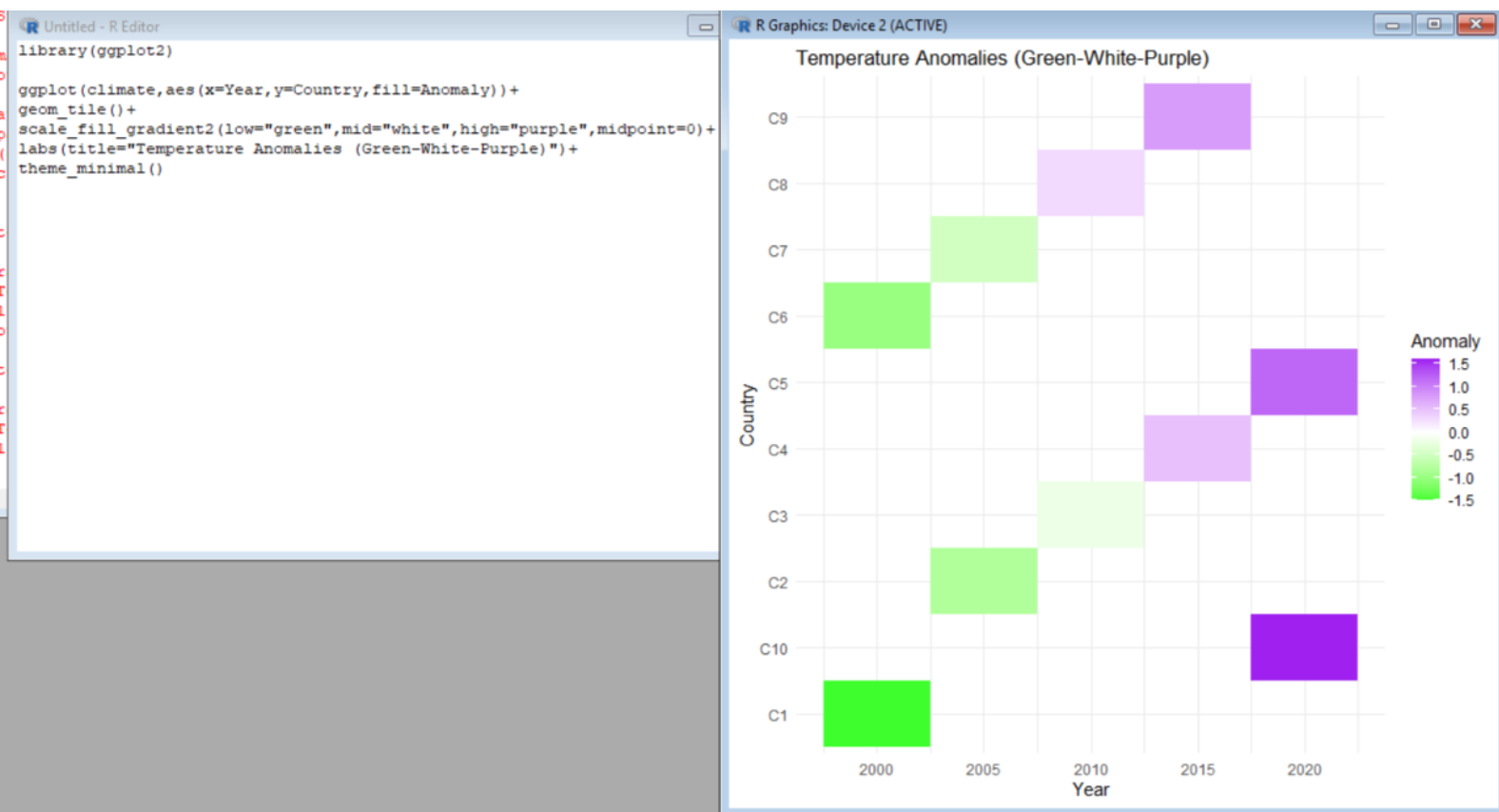
```
ggplot(climate,aes(x=Year,y=Country,fill=Anomaly))+
```

```
geom_tile()+
```

```
scale_fill_gradient2(low="green",mid="white",high="purple",midpoint=0)+
```

```
labs(title="Temperature Anomalies (Green-White-Purple)")+
```

```
theme_minimal()
```



Palette 3 – Cyan to Orange

```
library(ggplot2)
```

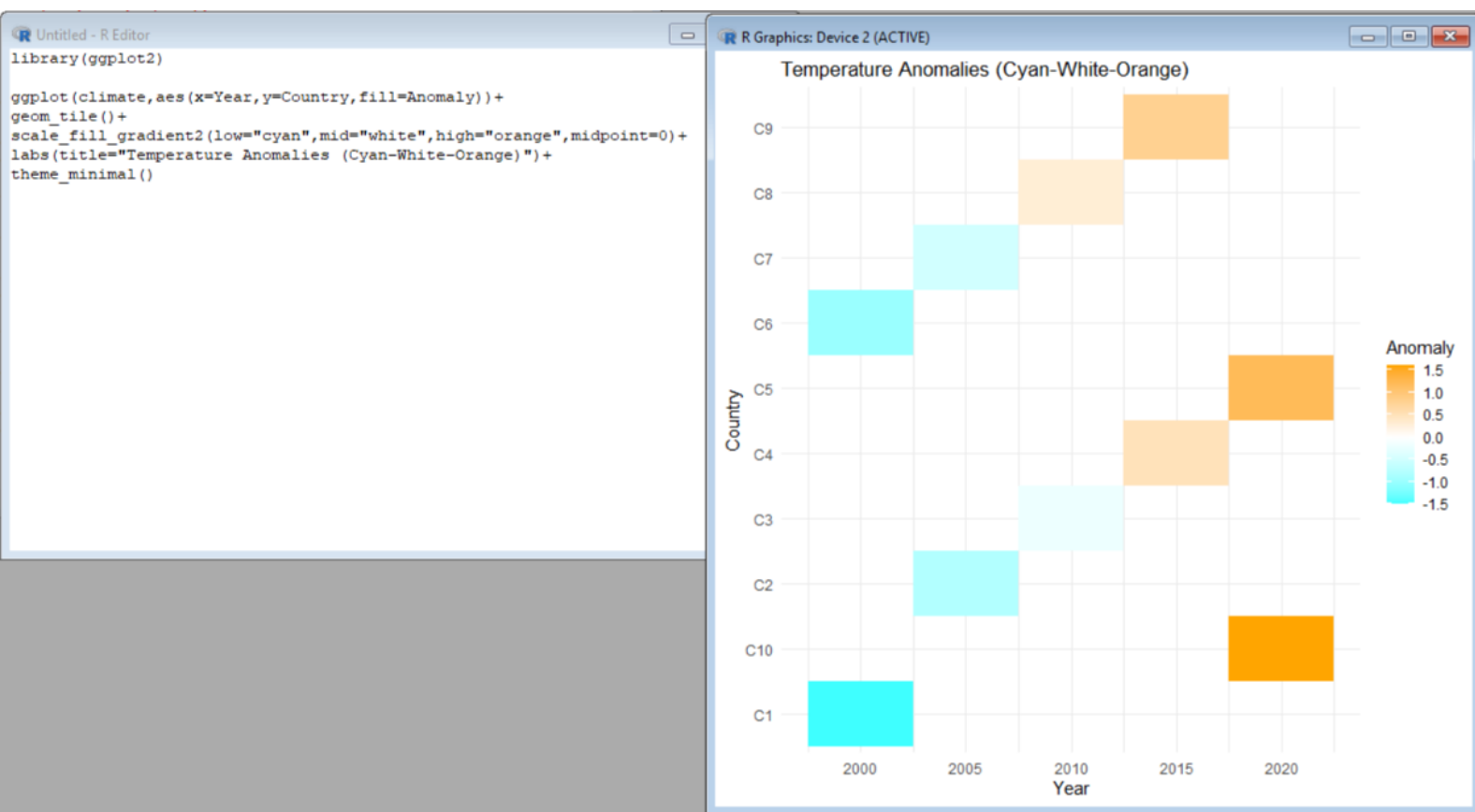
```
ggplot(climate,aes(x=Year,y=Country,fill=Anomaly))+
```

```
geom_tile()+
```

```
scale_fill_gradient2(low="cyan",mid="white",high="orange",midpoint=0)+
```

```
labs(title="Temperature Anomalies (Cyan-White-Orange)")+
```

```
theme_minimal()
```



6: E-Commerce Sales Analytics

```
library(ggplot2)
```

```
sales <- data.frame(
```

```
  Category=c("Electronics","Clothing","Furniture","Sports","Books"),
```

```
  Region=c("North","South","East","West","Central"),
```

```
  Sales=c(250,180,300,200,150)
```

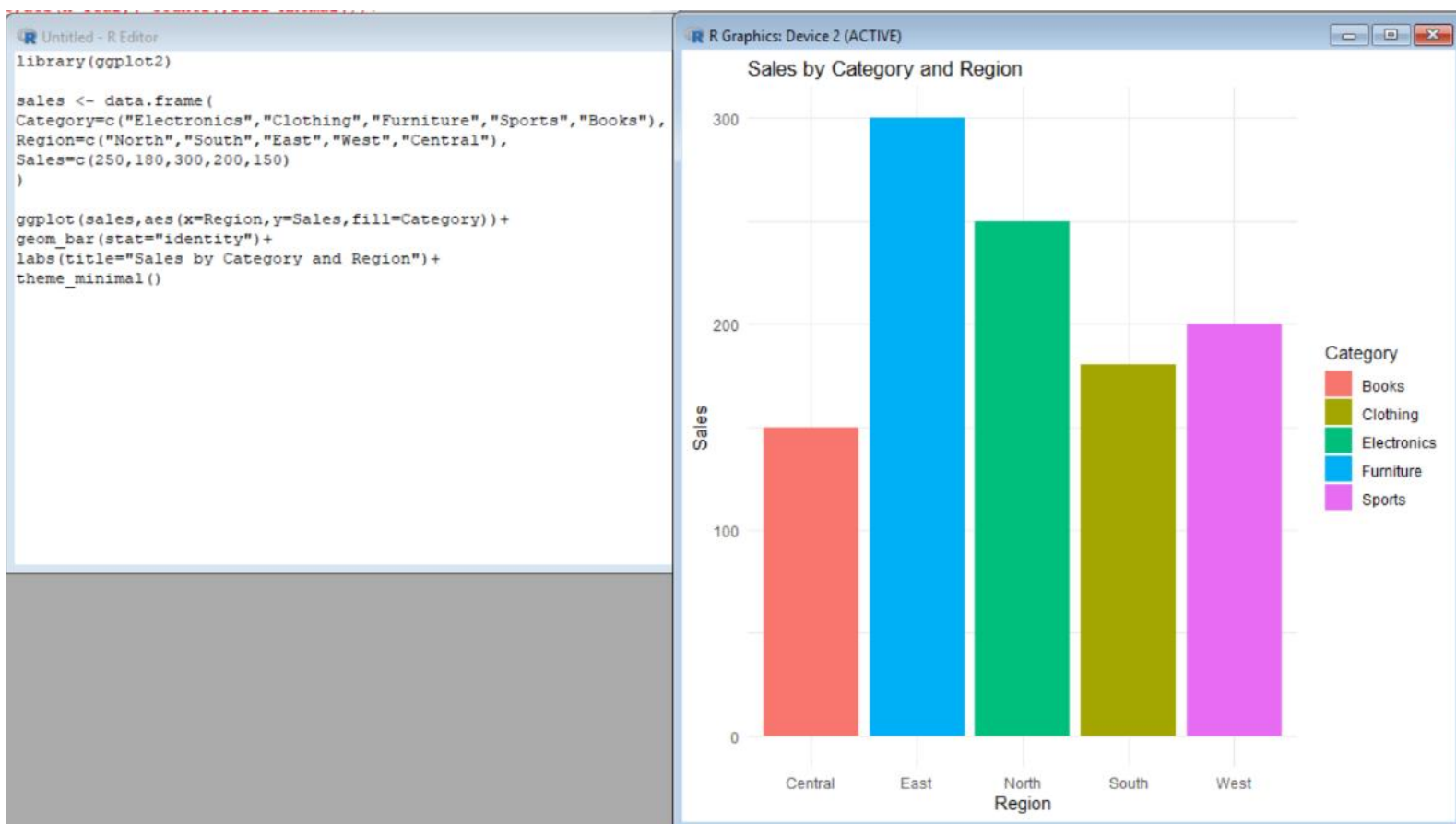
```
)
```

```
ggplot(sales,aes(x=Region,y=Sales,fill=Category))+
```

```
geom_bar(stat="identity")+
```

```
labs(title="Sales by Category and Region")+
```

```
theme_minimal()
```



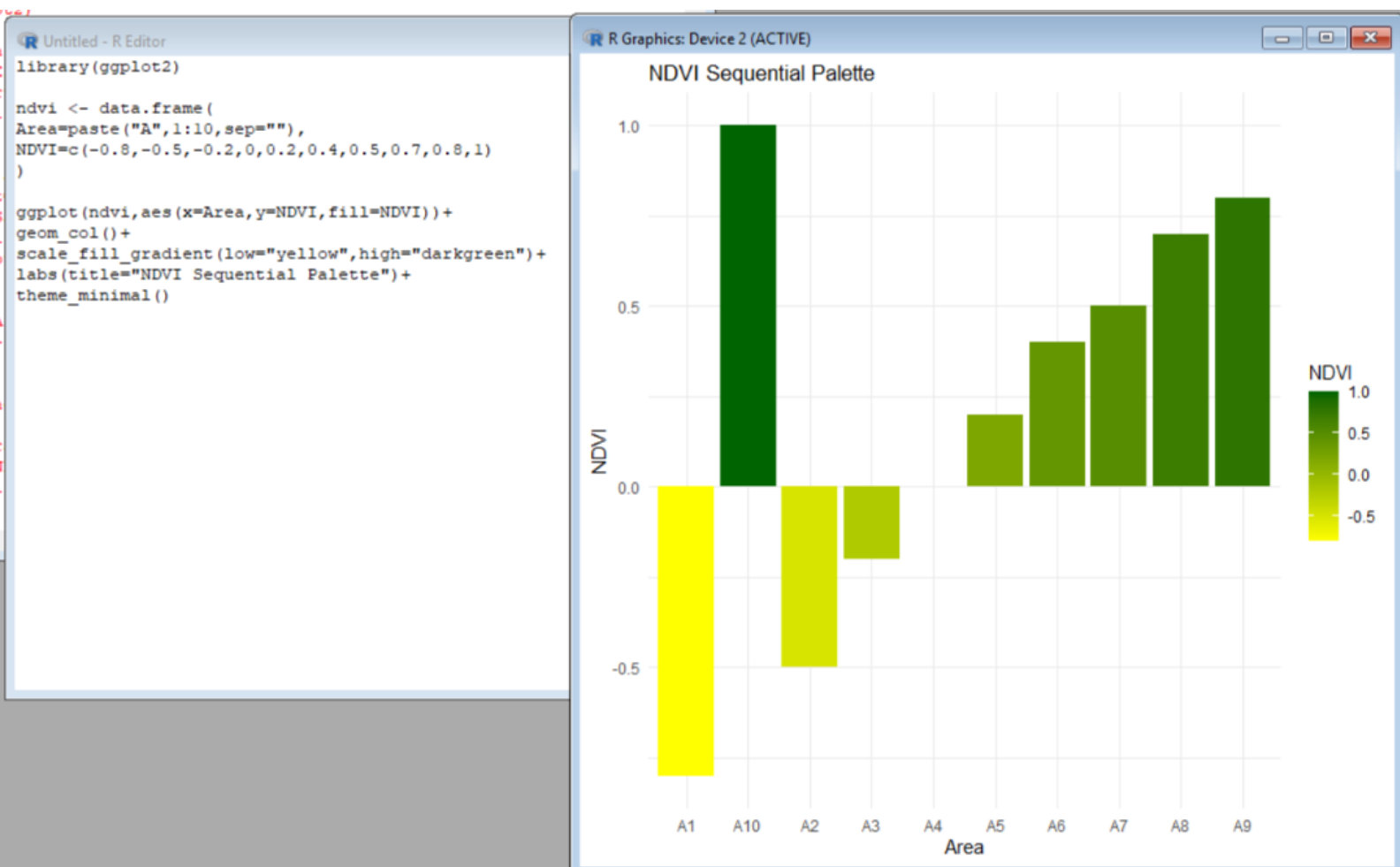
7: Satellite Vegetation Analysis

Sequential Palette

```
library(ggplot2)

ndvi <- data.frame(
  Area=paste("A",1:10,sep=""),
  NDVI=c(-0.8,-0.5,-0.2,0,0.2,0.4,0.5,0.7,0.8,1)
)

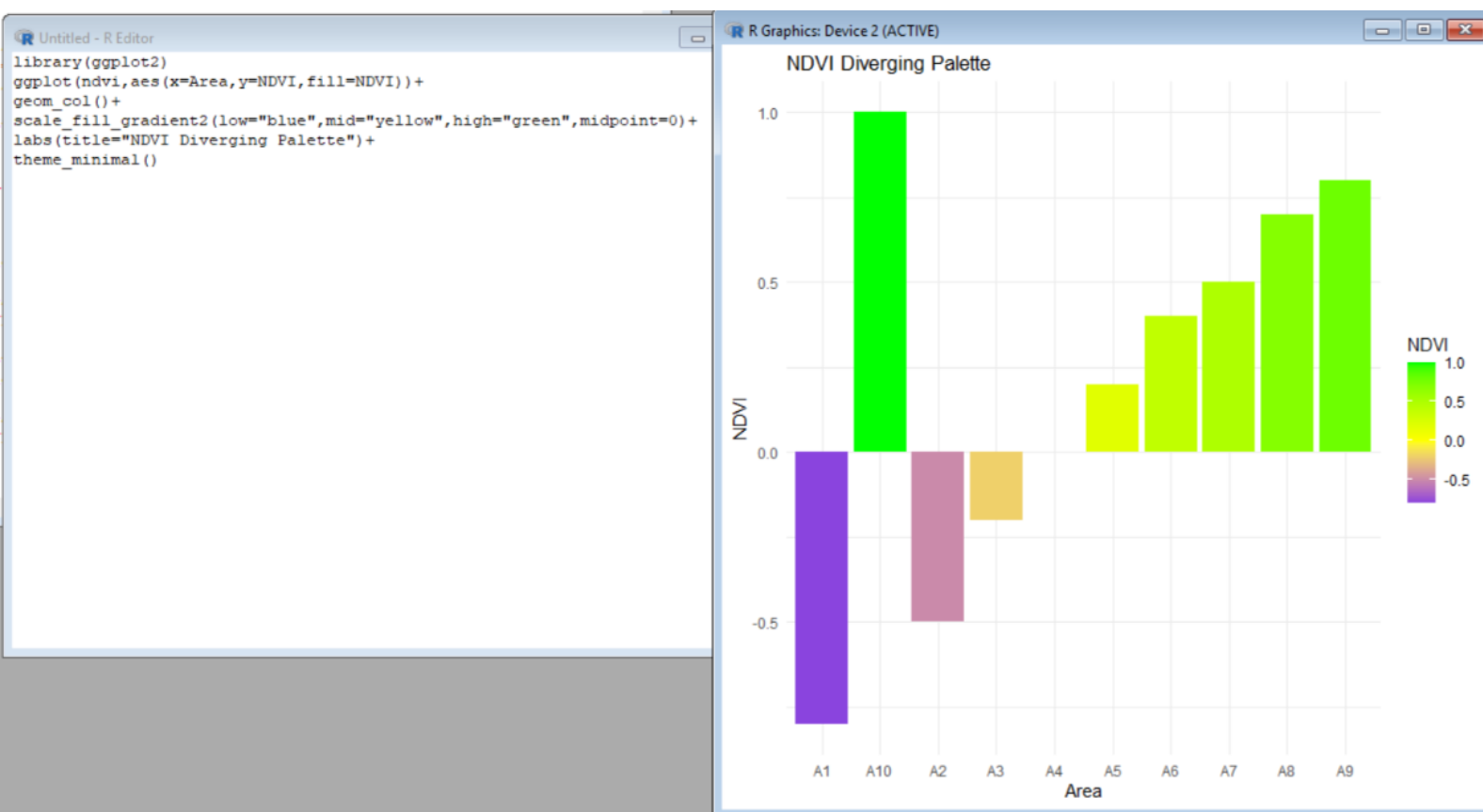
ggplot(ndvi,aes(x=Area,y=NDVI,fill=NDVI))+
  geom_col()+
  scale_fill_gradient(low="yellow",high="darkgreen")+
  labs(title="NDVI Sequential Palette")+
  theme_minimal()
```



Diverging Palette

```
library(ggplot2)

ggplot(ndvi,aes(x=Area,y=NDVI,fill=NDVI))+
  geom_col()+
  scale_fill_gradient2(low="blue",mid="yellow",high="green",midpoint=0)+
  labs(title="NDVI Diverging Palette")+
  theme_minimal()
```



8. Cryptocurrency Market Volatility

Cartesian Plot

```
library(ggplot2)
```

```
crypto <- data.frame(
```

```
Day=1:5,
```

```
Bitcoin=c(90000,92000,94000,96000,98000),
```

```
Ethereum=c(3000,3200,3100,3300,3400)
```

```
)
```

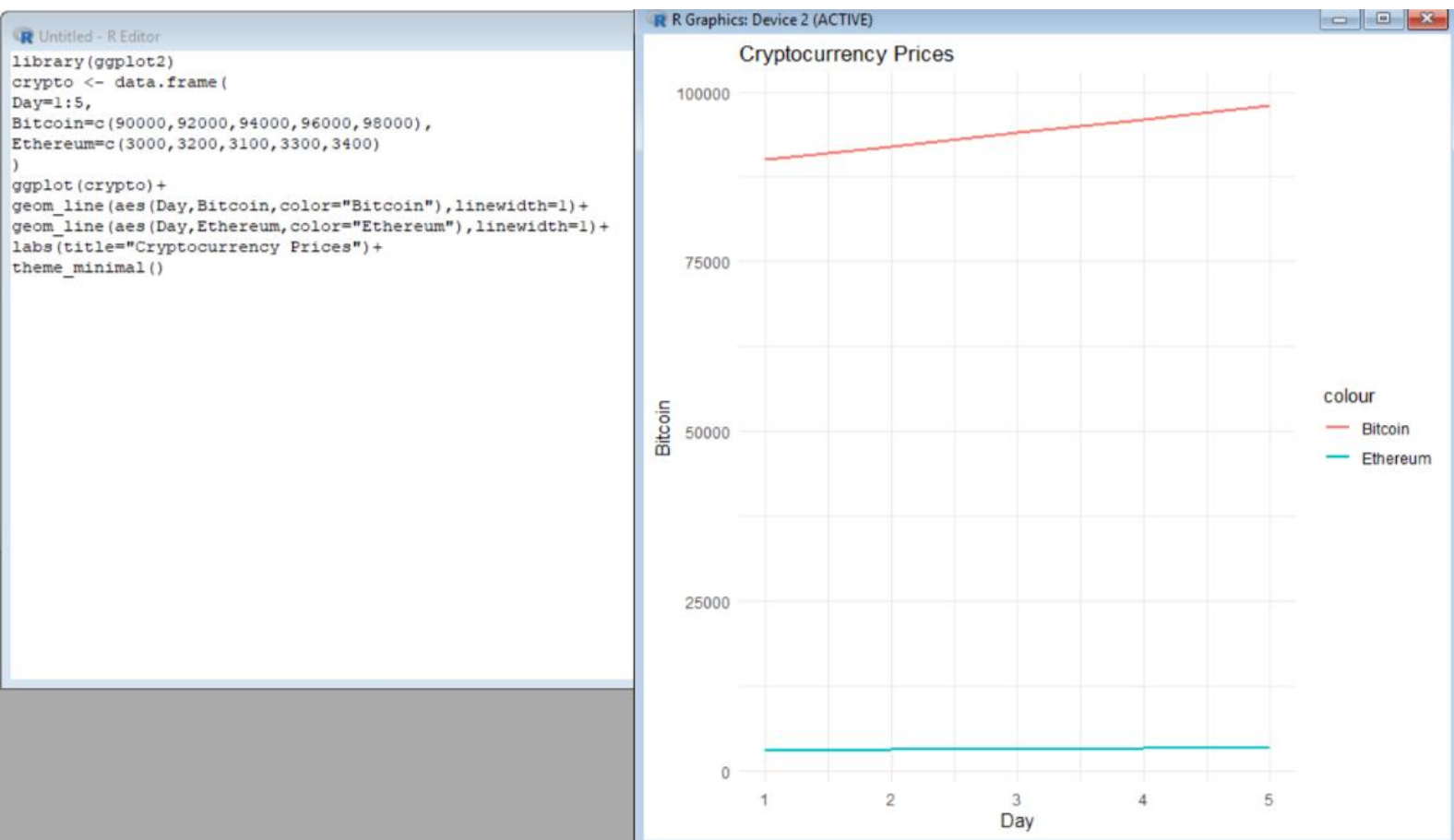
```
ggplot(crypto)+
```

```
geom_line(aes(Day,Bitcoin,color="Bitcoin"),linewidth=1)+
```

```
geom_line(aes(Day,Ethereum,color="Ethereum"),linewidth=1)+
```

```
labs(title="Cryptocurrency Prices")+
```

theme_minimal()



Logarithmic Axis

```
library(ggplot2)
```

```
ggplot(crypto)+
```

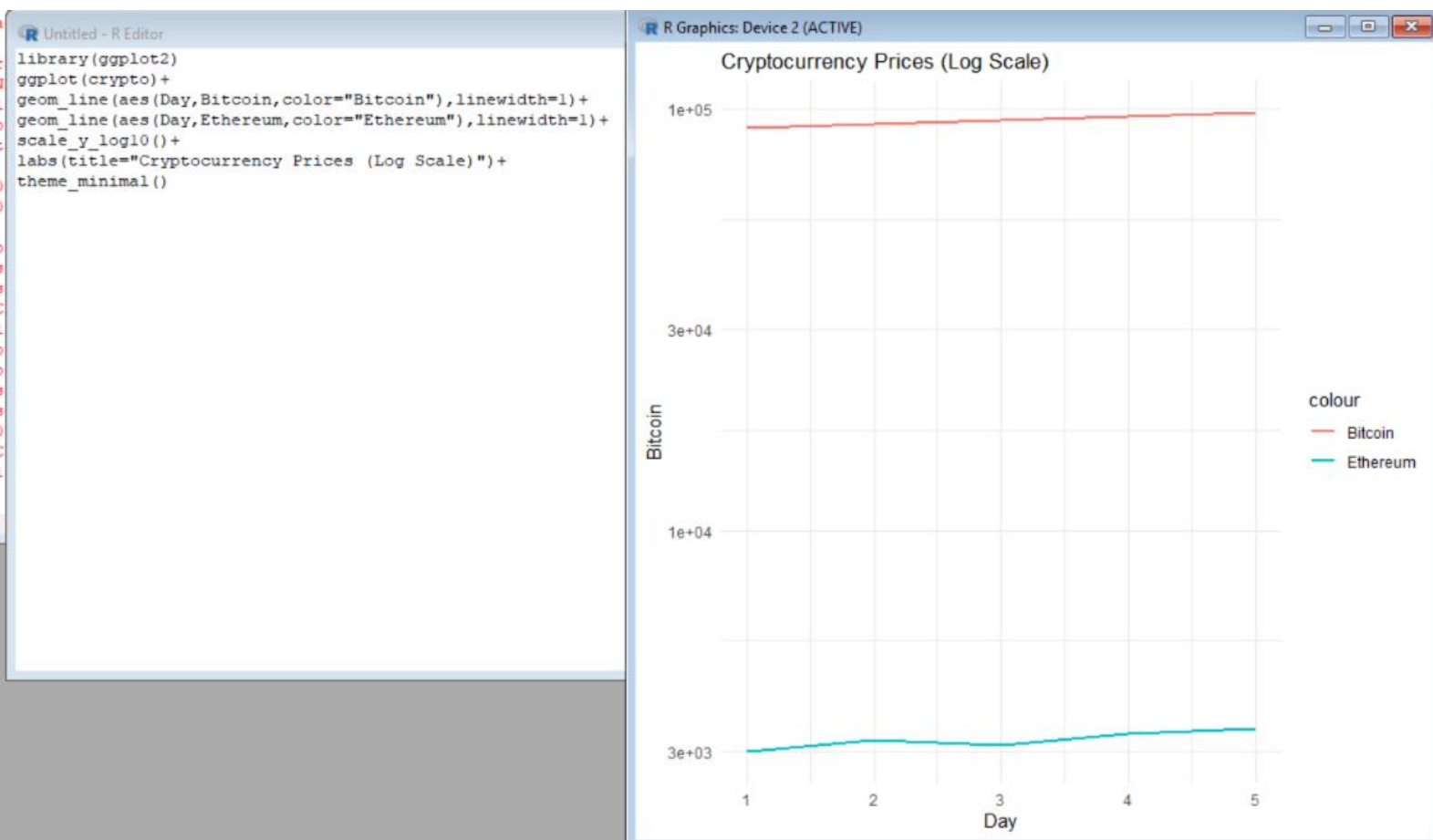
```
  geom_line(aes(Day, Bitcoin, color="Bitcoin"), linewidth=1)+
```

```
  geom_line(aes(Day, Ethereum, color="Ethereum"), linewidth=1)+
```

```
  scale_y_log10()+
```

```
  labs(title="Cryptocurrency Prices (Log Scale)")+
```

```
  theme_minimal()
```



9: Marathon Runner Performance

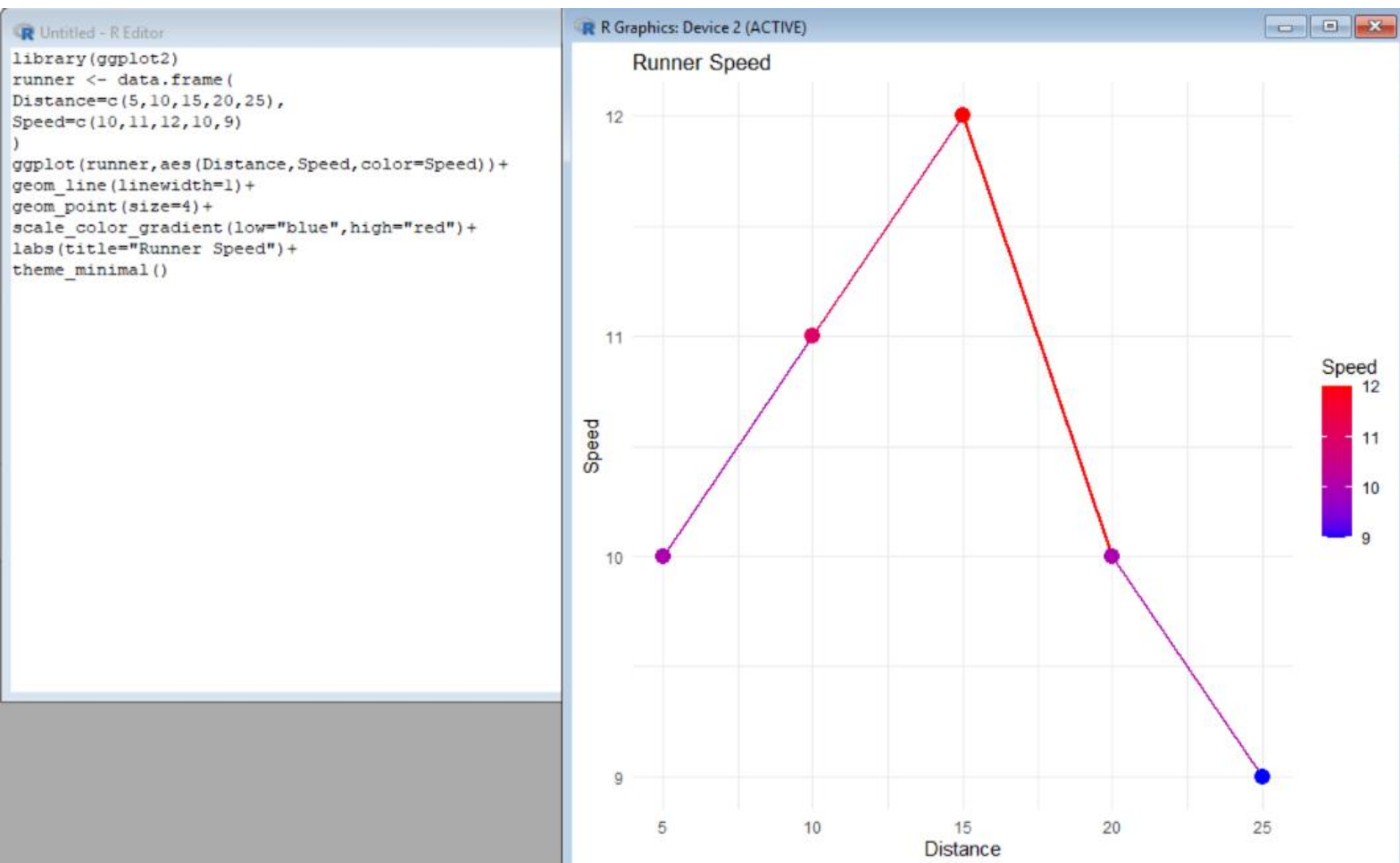
Cartesian Plot

```
library(ggplot2)

runner <- data.frame(
  Distance=c(5,10,15,20,25),
  Speed=c(10,11,12,10,9)
)

ggplot(runner,aes(Distance,Speed,color=Speed))+
  geom_line(linewidth=1)+
  geom_point(size=4)+
  scale_color_gradient(low="blue",high="red")+
  labs(title="Runner Speed")+
  theme_minimal()
```


theme_minimal()



Polar Plot

```
library(ggplot2)
```

```
runner2 <- data.frame(
```

```
  Direction=c("N","NE","E","SE","S","SW","W","NW"),
```

```
  Speed=c(10,11,12,10,9,11,10,9)
```

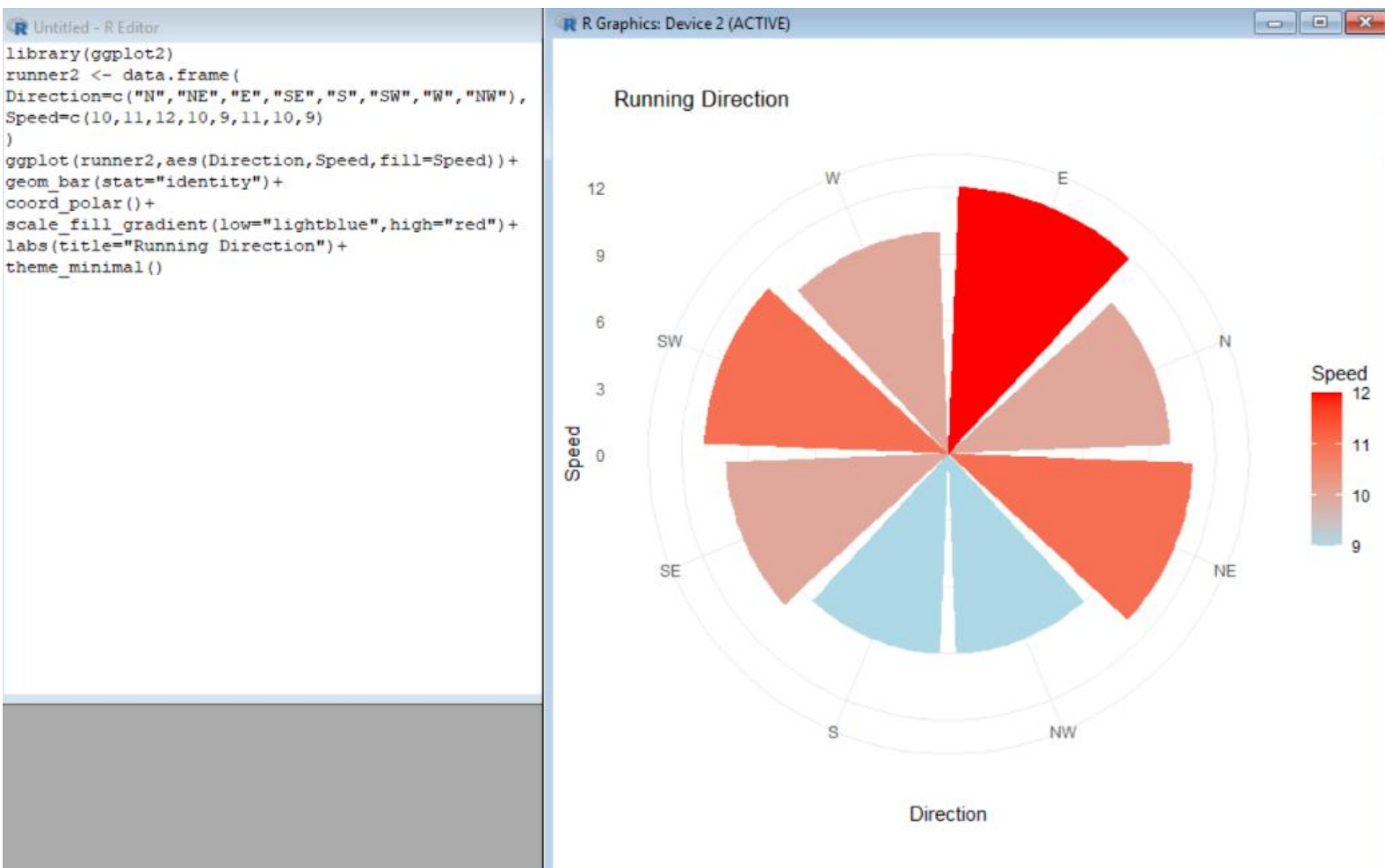
```
)
```

```
ggplot(runner2,aes(Direction,Speed,fill=Speed))+
```

```
geom_bar(stat="identity")+
```

```
coord_polar()+
```

```
scale_fill_gradient(low="lightblue",high="red")+  
labs(title="Running Direction")+  
theme_minimal()
```



10: COVID-19 Vaccination Dashboard

Version 1 (General Public)

```
library(ggplot2)
vaccine <- data.frame(
  District=c("A","B","C","D","E"),
  Rate=c(60,75,90,55,82)
```

```
)

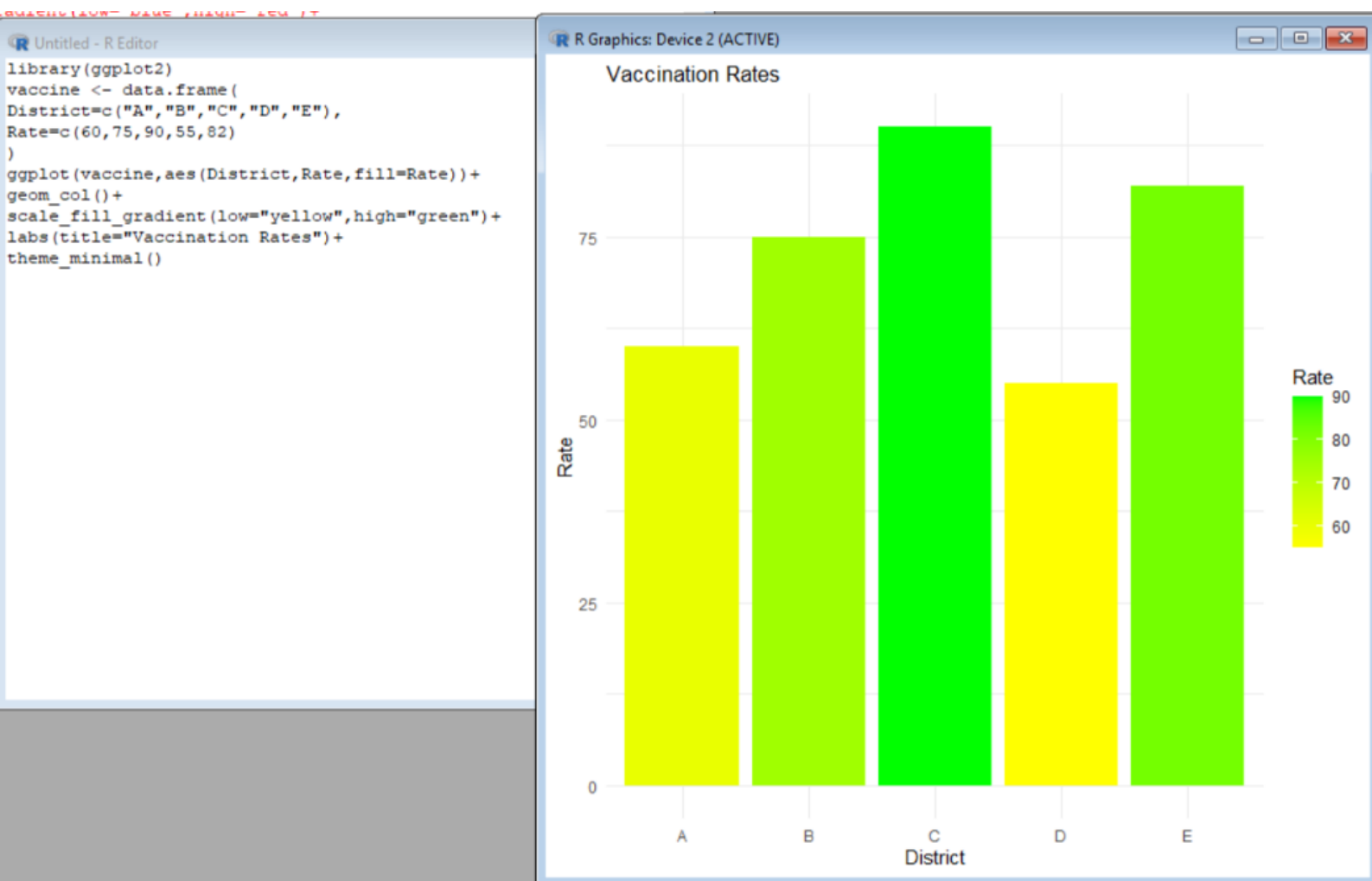
ggplot(vaccine,aes(District,Rate,fill=Rate))+

geom_col()+

scale_fill_gradient(low="yellow",high="green")+

labs(title="Vaccination Rates")+

theme_minimal()
```



Version 2 (Policymakers)

```
library(ggplot2)

vaccine$Status <- ifelse(vaccine$Rate>=80,"High","Low")

ggplot(vaccine,aes(District,Rate,fill=Status))+

geom_col()+
```

```
scale_fill_manual(values=c("High"="green","Low"="red"))+  
labs(title="Vaccination Status")+  
theme_minimal()
```

